

AI as an Adversarial Reader

Auditing Code & Papers Before You Submit

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What I'll Show You

One real project — a remote-education paper I've been trying to publish — and two ways I used AI on it:

1. Audit the code

Turn 3 years of accumulated do-files into a clean, reproducible replication package — and find bugs along the way.

2. Audit the paper

Adversarial referee-style review *before* resubmitting — then check it against the real referee reports.

The theme:

AI as a fast, tireless **adversarial reader** of work you've already done.

You stay the expert. It does the mechanical reading and cross-checking — at a scale and patience you can't match.

Erkmen follows: AI at the *front* end — brainstorming and pressure-testing ideas before you've written anything.

Part 1

Auditing a Replication Package

Bangladesh Remote Education RCT

Before: 3 Years of Accumulated Code

```
02_Do files/  
01_master_5th June 2021.do  
03_vargen_07sep.do  
03_vargen.do  
05_endline balance_eb.do  
05_endline balance.do  
09_attrition_8 July_e.do  
09_attrition_8 July.do  
...  
Archive/ (18 files)  
WB Submission Files/ (14 copies)
```

The problems:

- ▶ **51 do-files** across 4 subdirectories
- ▶ Master file from **June 2021** — 3 years stale
- ▶ Date-in-filename: `_07sep`, `_02July1130am`
- ▶ Author-in-filename: `_eb`, `_e`
- ▶ Hardcoded paths — only runs on my machine
- ▶ No documentation, no version control

This is **extremely common** in economics. You probably have a project that looks like this.

After: One Clean, Reproducible Pipeline

Dimension	Before	After
Do-files	51	8
Master file	2021	Current
Config system	None	Template
Version control	None	Git
Documentation	None	Full
"Which version?"	Constant	Zero

86 old variants preserved in Archive/ — nothing deleted, fully reversible.

```
bd-remote-ed/  
  
code/00_master_repo.do  
  
code/07_Dofiles_academic/  
_globals.do, 02_datagen.do,  
03_vargen.do (THE one), ...  
  
Archive/ (86 old)  
  
config/config_local_template.do  
  
docs/  justfile  README.md
```

Time: ~2 Claude Code sessions (a few hours). **By hand:** days, minimum.

The Bug It Found: A Typo Hiding in Plain Sight

🔍 The \$flags variable list

\$flags = 17 missing-value indicators used as controls in **every regression**. Defined independently in 5 scripts.

3 of 5 scripts (wrong):

```
... _f_c_engaged _f_c_health  
    _c_child_work
```

Missing the `_f_` prefix



Correct:

```
... _f_c_engaged _f_c_health  
    _f_c_child_work
```

Result: `_c_child_work` appeared *twice* in regressions; the real missing indicator was left out. Survived 10+ versions across 2+ years. No error, no crash — just quietly wrong. **How AI found it:** cross-referencing every global flags definition across all 51 files.

Part 2

Auditing the Paper

Can AI Referee Your Own Work?

What Is /econ-audit?

Q An Adversarial Review Skill

A reusable skill that reads a paper and acts as a **skeptical referee**.

- ▶ Reads the full paper (LaTeX, tables, appendix)
- ▶ Checks identification, power, measurement
- ▶ Produces a structured audit report
- ▶ Takes ~60 seconds

```
$ npx skills add eabeam/econ-skills \  
-skill econ-audit
```

What it checks:

1. **Identification:** Is the causal claim valid?
2. **Statistics:** MHT, power, attrition
3. **Measurement:** Are variables measured well?
4. **Methodology:** Design consistency
5. **Presentation:** Framing, length

Running It on the Paper: 5 Critical, 8 Important, 6 Minor

```
$ claude
/econ-audit main.tex
```

Output shown is from a saved run — AI output varies between runs, so I pin the version I'm reasoning about.

Three of the critical findings:



Critical

- ▶ **A1 — Identification:** the mediation claim (info → tutoring → learning) is not identified.
- ▶ **B1–B3 — Statistics:** no MHT adjustment across domains; headline effect below the MDE; underpowered heterogeneity drives the inequality story.
- ▶ **C1 — Measurement:** differential attrition in the learning sample ($p < 0.001$); suggested Lee bounds.

AI Audit vs. Real Referees: 7 of 10 in ~60 Seconds

Referee Criticism	AI?	Notes
High attrition & differential attrition	✓	Flagged as Critical; suggested Lee bounds
Results don't tell a coherent story	⦿	Caught ID problem, missed internal contradictions
Theoretical model not useful	✓	Flagged — but recommended <i>including</i> it
Short intervention duration (4–8 weeks)	✗	Requires field knowledge
Noisy learning measures (8-item phone test)	✓	Strong match; flagged IRT minimums
Paper too long and hard to follow	✓	Suggested cutting 20–30%
Multiple hypothesis testing issues	✓	Aggressive; added winner's curse framing
Limited external validity / COVID fatigue	⦿	Partial; missed publication landscape
Deviations from pre-analysis plan	✓	Caught the spirit, not the specifics
Complex randomization hard to interpret	✓	Strong match

5 full matches, 2 partial, 1 opposite recommendation, 2 misses.
11 referee reports across 8 journals over 4 years — vs. 60 seconds of AI.

What the AI Missed — And Why It Matters

✕ The Misses

- ▶ **Short duration:** 4–8 weeks is unusually short for an education RCT. Requires knowing field norms.
- ▶ **Internal contradictions:** the data arm raised tutoring *more* than the info arm, yet only info improved learning. Referees caught this; AI didn't.
- ▶ **COVID fatigue:** “we’ve seen too many of these.” AI can’t read the publication landscape.

The opposite advice:

The AI recommended *including* the conceptual model. Every referee wanted it **cut**. Sometimes AI gives exactly the wrong recommendation — because it lacks field conventions.

Pattern:

AI reliably catches **statistical / econometric** issues.

AI misses **field knowledge, within-paper logic checks, and publication strategy**.

Takeaway: A Very Good Pre-Submission Check

✓ What It Replaces

- ▶ The first 80% of what a careful internal seminar discussant would say
- ▶ Catches the systematic stuff: identification, power, MHT, attrition
- ▶ In one minute, not one month

What it doesn't replace:

- ▶ Field-specific intuition
- ▶ “Does this make sense given what we know about Bangladesh?”
- ▶ Publication strategy
- ▶ Colleagues who know your work

Code audit: it found bugs I hadn't found in two years. That alone justified the time.

Paper audit: run it before you send a paper out. It's free, and it catches the embarrassing stuff.

Try It — And Over to Erkmen



What to Try First

1. Install `/econ-audit`, run it on your next submission — before the referees do.
2. Point Claude Code at a messy project folder and ask it to audit the structure.

Same principle both times: **AI does the mechanical reading; you keep the judgment.**



Resources

- ▶ Econ skills:
`github.com/eabeam/econ-skills`
- ▶ Skills library:
`skills.sh`
- ▶ Claude Code docs:
`docs.anthropic.com/en/docs/claude-code`

Next: Erkmen on using AI to **brainstorm and pressure-test research ideas.**